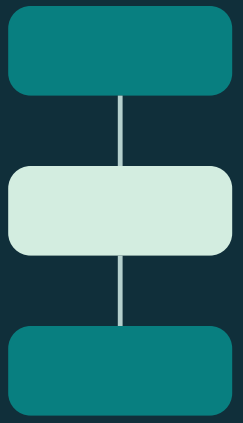
Aqion Relay



Recover supply.
Protect operating cash.

A practical recovery copilot for UAE SME distributors.

LIVE MVP https://duignytemvp.aqionlabs.workers.dev

GITHUB SOURCE https://github.com/Jaferdecodesai/DulgnyteMVP_Aqionlabs

A shipment slips.
Two commitments collide.

CUSTOMER COMMITMENT

Keep customer orders moving

A delayed inbound shipment creates an immediate gap in stock availability.

CASH COMMITMENT

Keep the business operating

A replacement purchase must leave room for existing payments and the owner’s cash floor.

Which recovery purchase is feasible — and worth approving?

Initial wedge: non-perishable packaging and general trading distributors; one SKU over 30 days.

One focused decision.
A strong fit with existing SME tools.

01 / RELEVANT

Disruption to decision

Stress-test shipment delay, demand and freight in one owner workflow.

02 / COMPLEMENTARY

Works beside the ERP

Read a normalized snapshot; return a reviewable recovery draft.

03 / DISTINCTIVE FOCUS

Cash-aware recovery

Compare service impact and minimum daily cash before approval.

Positioning: a recovery decision layer for the tools SMEs already use.

From disruption to an auditable draft.

1

Read

Stock, sales, scheduled cash and supplier terms.

2

Stress-test

Delay, demand and freight under owner-set limits.

3

Compare

Fulfilment, stockouts, spend and daily cash.

4

Review

Quantities, source freshness and the cash floor.

5

Approve

Save an unsent draft and export its evidence.

Every saved draft is bound to the reviewed source and scenario. The public MVP sends no order or payment.

A recovery choice you can inspect.

SYNTHETIC SIMULATION

PROJECTED UNITS FULFILLED



372 additional units served in this scenario

RECOVERY PURCHASE

AED 7,887.50

400 units · arrival day 4

AED 28,112.50

minimum cash vs AED 24,000 floor

Synthetic packaging fixture · 7-day delay · demand +20% · freight +25% · 130 evaluated combinations

Forecast numerically. Decide within hard constraints.

01 / FITTED ML

Forecast with a visible comparator

Weekday + trend ridge regression versus seasonal naive; labelled fallback for sparse history.

02 / NUMERICAL ENGINE

Simulate stock and daily cash

Bounded supplier combinations. MOQ, pack, cap, lead time, quote validity and cash floor.

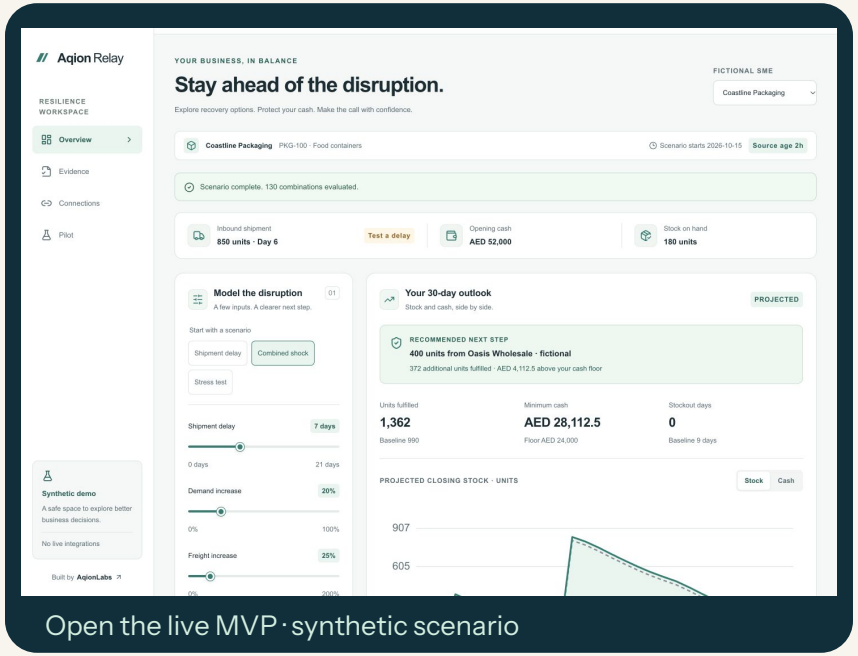
03 / HUMAN AUTHORITY

Explain, review and confirm

Traceable assumptions and version checks. The owner confirms the unsent recovery draft.

Forecast output informs a scenario; constraint checks and explicit owner approval control the action.

A mobile workbench, with a complete decision trail.



Open the live MVP · synthetic scenario

01 **Compare clearly**

Stock and cash curves, daily data and feasible options.

02 **Review deliberately**

Confirmation, stale-source blocks and duplicate protection.

03 **Keep the evidence**

Persisted drafts, readable audit history and portable JSON.

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One deployment.
Persistent, isolated demo workspaces.

MOBILE WEB

Next / React UI

Responsive workbench
Accessible table alternative

CLOUDFLARE WORKER

Assets + API routing

Serve the React workbench
Route the workspace request

DURABLE OBJECT

Engine + SQLite

Forecast, simulate, approve
Atomic state + draft + audit

Signed session + CSRF | **Bounded input + safe errors** | **Version-bound approval** | **Human confirmation**

SYSTEM DESIGN https://github.com/Jaferdecodesai/DulgnyteMVP_Aqionlabs/blob/main/Aqion-Relay-System-Design.docx

Live scope: synthetic data and unsent drafts. Enterprise identity, ERP credentials and live writes enter through pilot gates.

Read first. Reconcile every write. Then expand through du.

WEEKS 1–2

Map + authorize

Consented snapshots, terms and source IDs

Gate: data + security sign-off

WEEKS 3–4

Connect read-only

Zoho stock, sales and scheduled cash mapping

Gate: source reconciliation

WEEKS 5–6

Draft + reconcile

Owner approval, ERP outbox and idempotency

Gate: sandbox failure tests

WEEKS 7–8

Enable distribution

Agreed du interfaces, support and monitoring

Gate: sponsor acceptance

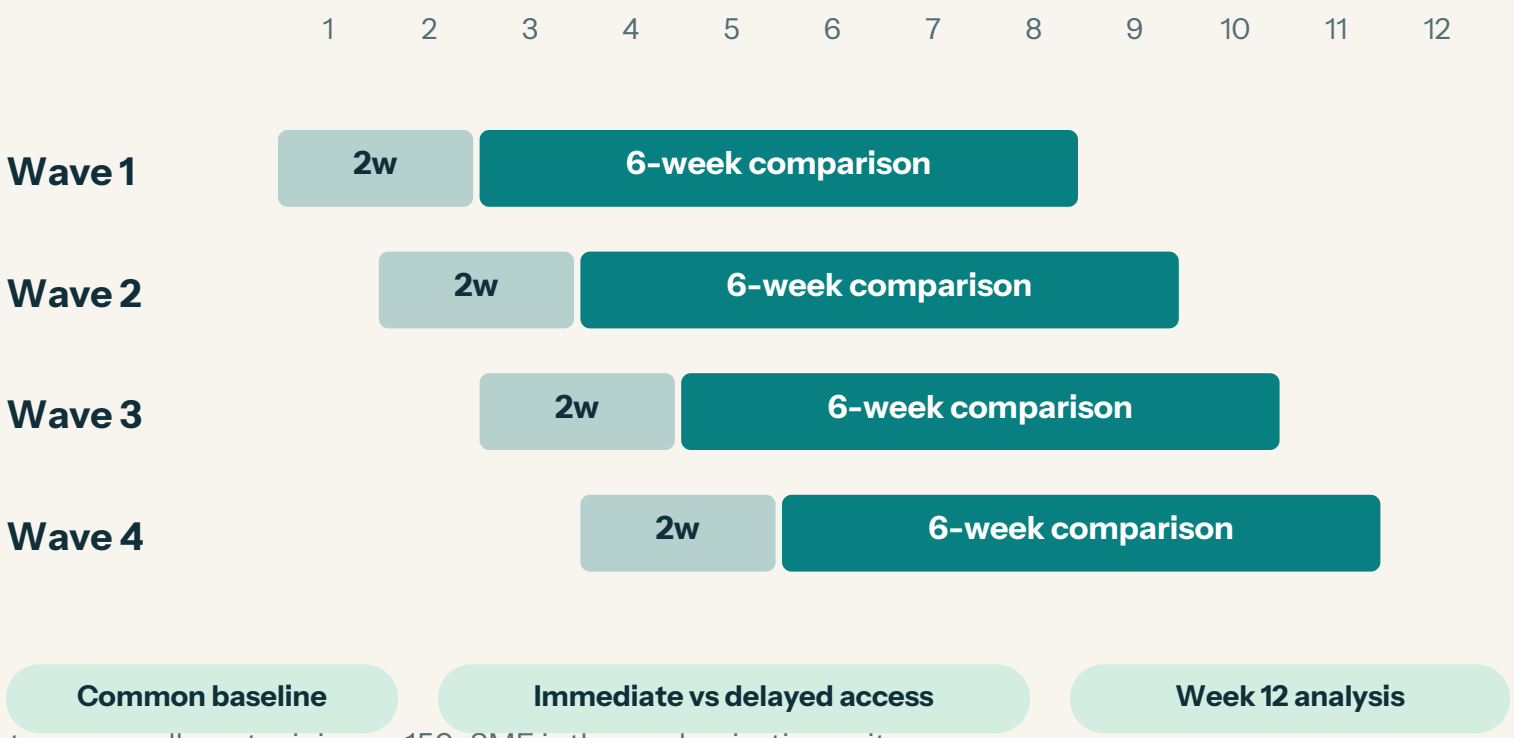
Accountability: AqionLabs builds · SME owner validates · du sponsors integration and acceptance

160 SMEs. Four waves. A credible comparison.

160

Enrollment target

4 × 40
20 immediate +
20 delayed per wave



Proposed 12-week pilot after integration acceptance · enrollment minimum 150 · SME is the randomization unit

Measure better decisions.
Keep operational guardrails absolute.

40%

Faster feasible planning

Primary target: common disruption drill to owner-approved, rubric-passing plan.

90%

Feasible-plan rate

Target share of assigned drills with an approved plan passing blinded scoring.

70%

Weekly workflow use

Target share of SMEs with access reviewing at least one scenario each week.

STOP RULES / Projected cash-floor breach · unauthorized write · duplicate external effect

Pilot targets, not measured benefits. Pre-register analysis; report effect sizes, confidence intervals, attrition and deviations.

A narrow wedge.
A repeatable route through du.

160

Pilot cohort target

Validate decision value, source quality and support demand.

1,000

Controlled rollout

Standardize connectors and onboarding; enforce cost limits.

40,000+

Scale ambition

Partition workloads; prove peak load, isolation and restore.

Commercial offer Per-SME subscription + agreed connector onboarding, distributed through du.

Scale sequence and subscription are proposals. Expansion follows measured adoption, unit cost, isolation and recovery acceptance.

Make the safe action the easiest action.

Input quality	Freshness, required fields and valid supplier terms	**Block stale or incomplete approval**
Execution integrity	Human confirmation, version checks and idempotency	**ERP outbox + reconciliation gate**
Data protection	Isolated demo state; signed session + CSRF	**Pilot IAM, approved processing and restore test**
Model reliability	Comparator, fallback and visible assumptions	**Monitor forecast error and rejected scenarios**

Accountable operations: monitor exceptions · pause writes on a material breach · reconcile before resuming

Launch the du recovery pilot.

A working product.
A measurable pilot.
A clear path to SME scale.

NEXT MILESTONE / Pilot charter + named owners

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01 **Sponsor + recruit**

160 consenting SMEs; minimum 150 enrolled.

02 **Connect + operate**

Named du technical owners and approved interfaces.

03 **Evaluate + expand**

Joint scorecard, delivery gates and cost ceiling.